

Solar Load Calculator

Google Colab Notebook -- Step-by-Step Guide

Electricity Bill -> AI Extraction -> Solar Sizing -> Excel Report

FREE | No credit card | 5 minutes to set up

What you will be able to do:

- [+]** Upload any Indian electricity bill (PDF, JPG, or PNG)
- [+]** Let AI automatically read and extract all key numbers from the bill
- [+]** Get the correct solar system size, estimated savings, and ROI
- [+]** Download a formatted Excel report ready to share with customers

Supported Indian electricity bills:

MSEDCL / BESCO / TATA Power / CESC / Adani Electricity / and other utilities

AI model: Google Gemini 2.0 Flash (free tier -- no payment required)

All Steps at a Glance

Step	Action	What Happens
Step 1	Install packages	Run once per session -- installs the 3 required libraries
Step 2	Get a free API key	Create a free Gemini key at Google AI Studio (no card needed)
Step 3	Define functions	Run once to load the AI and calculation code into memory
Step 4	Upload your bill	Pick your PDF, JPG, or PNG -- AI reads and extracts all data
Step 5	Download Excel report	View the results and save the formatted spreadsheet

STEP 1 -- Install Required Packages

Run the first code cell. It installs three Python libraries needed by the notebook:

- `google-genai` -- Google Gemini AI -- reads and understands the electricity bill
- `pdfplumber` -- Extracts text from PDF bills
- `openpyxl` -- Builds the formatted Excel report

[TIP] Run this cell ONCE per Colab session. If you close and reopen Colab, run Step 1 again first.

[OK] When done you will see: [OK] All packages installed successfully.

STEP 2 -- Get Your Free Gemini API Key

The notebook uses Google Gemini AI to read the bill. You need a free API key. Here are the two parts:

Part A -- Get your key from Google AI Studio:

- 1. **Open your browser and visit:** `https://aistudio.google.com/app/apikey`
- 2. **Sign in with:** Any Google / Gmail account
- 3. **Click:** "Create API Key" -- choose any project or "Create in new project"
- 4. **Copy the key shown:** It starts with `AlzaSy...` (about 40 characters long)

Part B -- Save the key in Colab Secrets (keeps it private):

- 1. **In Colab:** Look for the key icon in the left sidebar and click it
- 2. **Click:** "+ Add new secret"
- 3. **Name:** Type exactly `GEMINI_API_KEY` (capitals, underscore, no spaces)
- 4. **Value:** Paste the key you copied in Part A
- 5. **Toggle:** Turn ON "Notebook access" so the notebook can read the key
- 6. **Run Step 2 cell:** You should see: [OK] Gemini API key loaded from Colab Secrets

[SECURE] Your key is stored securely in Colab and is NOT in the notebook file. You can safely share the `.ipynb` file with anyone.

STEP 3 -- Define All Functions

This cell is long but you do not need to read the code. Just click Run. It loads three functions:

<code>extract_bill_data()</code>	Sends your bill to Gemini AI and returns all key fields as structured data
<code>calculate_solar_recommendation()</code>	Calculates system size, savings, payback period, and CO2 reduction
<code>generate_excel()</code>	Creates a formatted colour-coded Excel spreadsheet with all results

[OK] When this cell finishes you will see: [OK] All functions ready.

STEP 4 -- Upload Your Electricity Bill

Run this cell -- a file upload button appears. Here is exactly what happens:

- | | |
|-------------------------|---|
| 1. File picker appears: | Click "Choose Files" and select your electricity bill from your computer. |
| 2. File is uploaded: | The filename and file size are shown below the cell. |
| 3. AI reads the bill: | The bill is sent to Gemini. Takes 5-15 seconds. You will see: "Sending bill to Gemini AI for extraction..." |
| 4. Calculations run: | System size, savings, payback period, and CO2 reduction are calculated instantly. |
| 5. Excel is prepared: | The report is built in memory. You will see: [Done] Run Step 5 to download. |

Accepted file formats:

- | | |
|-------------------------|---|
| <code>.pdf</code> | -- Text-based PDF downloaded from MSEDCL, BESCO, TATA Power, etc. |
| <code>.jpg/.jpeg</code> | -- Photo of the bill taken on your phone |
| <code>.png</code> | -- Screenshot of the bill from your computer or phone screen |

[TIP] If the PDF has no selectable text (scanned bill), upload a JPG/PNG screenshot instead. Bills downloaded from utility portals work perfectly as PDFs.

STEP 5 -- View Results and Download the Excel Report

Run this cell to print all the results on screen and automatically download the Excel file.

- | | |
|---------------------------|---|
| > Customer Information -- | Consumer name, number, meter number, distribution company, billing month, tariff category |
| > Electricity Usage -- | Units consumed (kWh), sanctioned load (kW), bill amount, cost per unit, daily usage |
| > Solar Recommendation -- | System size (kWp), monthly/annual savings, payback period, CO2 reduction, 25-year benefit |

[DOWNLOAD] The Excel file downloads automatically at the end. File name: solar_load_[ConsumerNumber].xlsx

Troubleshooting Common Errors

[!] API key not found <i>When you see: "Gemini API key not found"</i> Fix: Open Colab Secrets (key icon, left sidebar), add GEMINI_API_KEY, turn on Notebook access. Then re-run Step 2.
[!] PDF has no readable text <i>When you see: "This PDF appears to be scanned"</i> Fix: Your PDF is image-only. Take a screenshot or photo of the bill and upload it as a JPG or PNG instead.
[!] No results in Step 5 <i>When you see: Step 5 says "No results yet -- please run Step 4 first"</i> Fix: Run Step 4 first, upload a bill and wait for the Done message, then run Step 5.
[!] Module not found <i>When you see: "No module named google.genai" or similar</i> Fix: The Colab runtime was reset. Re-run Step 1 to reinstall packages, then re-run Steps 2 and 3.
[!] Function not defined <i>When you see: "name extract_bill_data is not defined"</i> Fix: Re-run Step 3 to reload the functions. This is needed after any Colab runtime restart.

What the Downloaded Excel File Contains

One sheet with four colour-coded sections:

Section 1	Customer Information	Consumer name, number, meter number, distribution company, billing month, tariff category
Section 2	Electricity Usage	Units consumed (kWh), sanctioned load (kW), total bill amount, cost per unit (Rs/kWh), daily usage
Section 3	Solar Recommendation	Recommended system size (kWp), monthly/annual savings, payback period (years), CO2 reduction
Section 4	Financial Summary (25 yrs)	Estimated system cost, total 25-year savings, and net benefit

Ready to submit your AI intern task?
Send the filled Excel output + a screenshot of the notebook running:
Email: pranay@energybae.in | Subject: AI Intern Task -- [Your Name] | WhatsApp: +91 9112233120